

AI-Powered Customer Support Analytics

Business Case — One-Page Summary

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8,469

Tickets Analyzed

32.70%

Resolution Rate

39.80%

Observed Dissatisfaction

39.64%

Predicted Risk (avg)

91.31%

Negative Sentiment

The Business Problem

Support teams need fast answers on ticket volume, resolution effectiveness, emerging complaint themes, and which tickets carry the highest risk of a dissatisfied customer — questions conventional status/priority reporting cannot answer because it ignores ticket text.

The Solution

An end-to-end pipeline turns 8,469 raw tickets into a validated DuckDB analytical layer using three AI signals — sentiment classification, BERTopic support-theme discovery, and a Gradient Boosting dissatisfaction-risk model — surfaced through a three-page Power BI dashboard (Executive Overview, Issue Analysis, AI Insights), fully cross-validated against SQL/Python with zero discrepancies.

Key Results

Metric	Result
CSAT-observed tickets	2,769 of 8,469
Dissatisfaction model (Gradient Boosting)	Recall 91.4% / Precision 39.9%
Sentiment & topic outputs	8,469/8,469 — 0 missing
Dashboard QA discrepancies	0 (cross-checked)

Observed vs. Predicted Dissatisfaction are distinct: the observed rate covers only CSAT-rated tickets; the predicted rate is the model's average risk score across all 8,469 tickets. They are never presented as interchangeable.

Engineering Decision: Scope Correction

The original target was escalation risk, but profiling found no defensible escalation label (no explicit field, ~49% invalid timestamp ordering, no reliable creation timestamp). Rather than fabricate one, the target was re-scoped to a CSAT-grounded **is_dissatisfied** outcome — preserving the original goal of proactive ticket triage on a target the data can actually support.

AI-Driven Insights

- **Sentiment** — pretrained NLP model scores all tickets (91.31% negative); a 20-ticket diagnostic check (50% agreement) flags known domain-mismatch limits.
- **Topics** — BERTopic surfaces 30 recurring themes (firmware, account access, connectivity, billing/refunds, hardware) from ticket text alone.
- **Dissatisfaction Risk** — trained only on the 2,769 CSAT-observed tickets, then scored across all 8,469; threshold (0.30) favors recall so more at-risk tickets surface for review, at the cost of more false positives.

Business Value by Stakeholder

Stakeholder	Value
Support Leads	Prioritize review via risk scores
Ops Managers	Monitor volume, resolution, dissatisfaction
CX Teams	Investigate sentiment & complaint themes
Product Teams	Spot recurring technical/service issues
Leadership	Executive pulse + AI-flagged risk areas

Responsible Use

AI outputs support human decisions — they do not replace them. Outputs must not be used to auto-reject customers, auto-deny refunds, or auto-classify dissatisfaction without review. **dissatisfaction_risk_score** is a prioritization signal, never a confirmed outcome, and is kept visually and analytically separate from observed CSAT throughout the SQL layer and dashboard.

Key Limitations

- Supervised model trained on ~33% of tickets (CSAT-observed only)
- General-domain sentiment model, not domain fine-tuned
- Threshold selected on evaluation set — needs independent validation
- No defensible ticket-creation timestamp in source data